

Guru Nanak Dev Engineering College, Ludhiana



Project Title: Student Grade Management System

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Introduction

The Student Grade Management System is a simple console-based C++ program that helps a teacher store and manage student records in an easy way. In schools, teachers need to keep track of many students and maintain their marks, totals and grades. This program shows how the same work can be done through a small computer application.

The program uses a Student class to store details such as the student's name, roll number, marks of different subjects, total marks, percentage and the final grade. Using a class keeps all information about one student together, which makes the program organized and simple to update. It also shows the basic idea of object-oriented programming where data and functions are grouped together.

An important feature of this project is file handling. Instead of keeping data only for the current run, the program saves all student records in a file called students.txt. Because of this, the information stays safe even after the program is closed. When the program starts again, it reads the saved records from the file and uses them. This is similar to how real school systems store student data for long-term use.

When the program runs, it shows a menu with different options. A teacher can add a new student and enter their marks. The program automatically calculates the total marks, average percentage and assigns a grade based on the percentage. The teacher can also view all student records or see the report card of a specific student using the roll number. If any information needs correction, the edit option allows changes to be made. The delete option removes a student record that is no longer needed. Every change is saved back to the file so that the data always remains correct.

Overall, the Student Grade Management System is a simple and useful project for understanding how C++ can handle real-life tasks. It explains file handling, classes, and menu-driven programs in an easy way.

Design (Modules, Functionality)

Program store one or multiple student records in a file and perform grade-related operations. The program uses a Student class and array Student s1[100] for storage while the program runs.. Data is saved in a simple text file students.txt. The program is menu-driven with options to add, view, edit, delete and exit.

1. Student Class

This is the core where a student's details and academic data are stored.

Data members

- int rollNo
- string name
- int marks[10]
- int total
- float average
- char grade
- int n

Member functions

1. void inputDetails()

- Takes roll number using cin.
- Reads full name using getline.
- Takes number of subjects (n).
- Calculates total and average = total / n.
- Assigns grade based on average.

2. void displayDetails()

- Prints student name and roll number.
- Shows total marks and average.
- Displays grade clearly.

3. int getRollNo()

- Simply returns the roll number.
- Used for searching and editing a student.

4. void SaveToFile(ofstream &out)

- Writes roll number to file.

- Writes name to next line.
- Writes 10 marks (fixed), even if the student had fewer subjects.
- Does NOT save n, which causes mismatch when reading later.

5. bool readFromFile(ifstream &in)

- Reads roll number from file.
- Reads name.
- Reads marks into the marks array.
- Recalculates total and average (using existing n value).
- Returns false when no more data is available.
- Limitation: n is not loaded from file.

6. void editDetails()

- Allows user to enter new name.
- Takes new n .
- Inputs updated marks.
- Recomputes total, average, and grade after editing.

2. Reading and Writing File Module (File Handling Functions)

Use a file students.txt to save all students or a single student depending on requirement.

Functions

a) SaveToFile(ofstream &out)

- Writes a single student's details (roll number, name, marks) to a file.
- Called in main() when saving all students

b) readFromFile(ifstream &in)

- Reads a single student's details from the file.
- Returns false if the end of the file is reached.
- Called in main() to load all students at the start:

3. Functional Modules

These functions implement user-facing operations.

a) inputDetails()

- To input a student's details (roll number, name, number of subjects, marks).
- Input marks for n subjects.
- Compute total, average, and grade.

b) displayDetails()

- To display a student's details in a formatted way.
- Shows name, roll number, total marks, average, and grade.

c) editDetails()

- To modify an existing student's name and marks.
- Input new name and marks
- Update total, average, and grade

d) displayStudent()

- Reads student by ID and calls show() to display details.

e) SaveToFile(ofstream &out)

- To save student details to a file (students.txt).
- Write roll number, name, and marks to file.

f) readFromFile(ifstream &in)

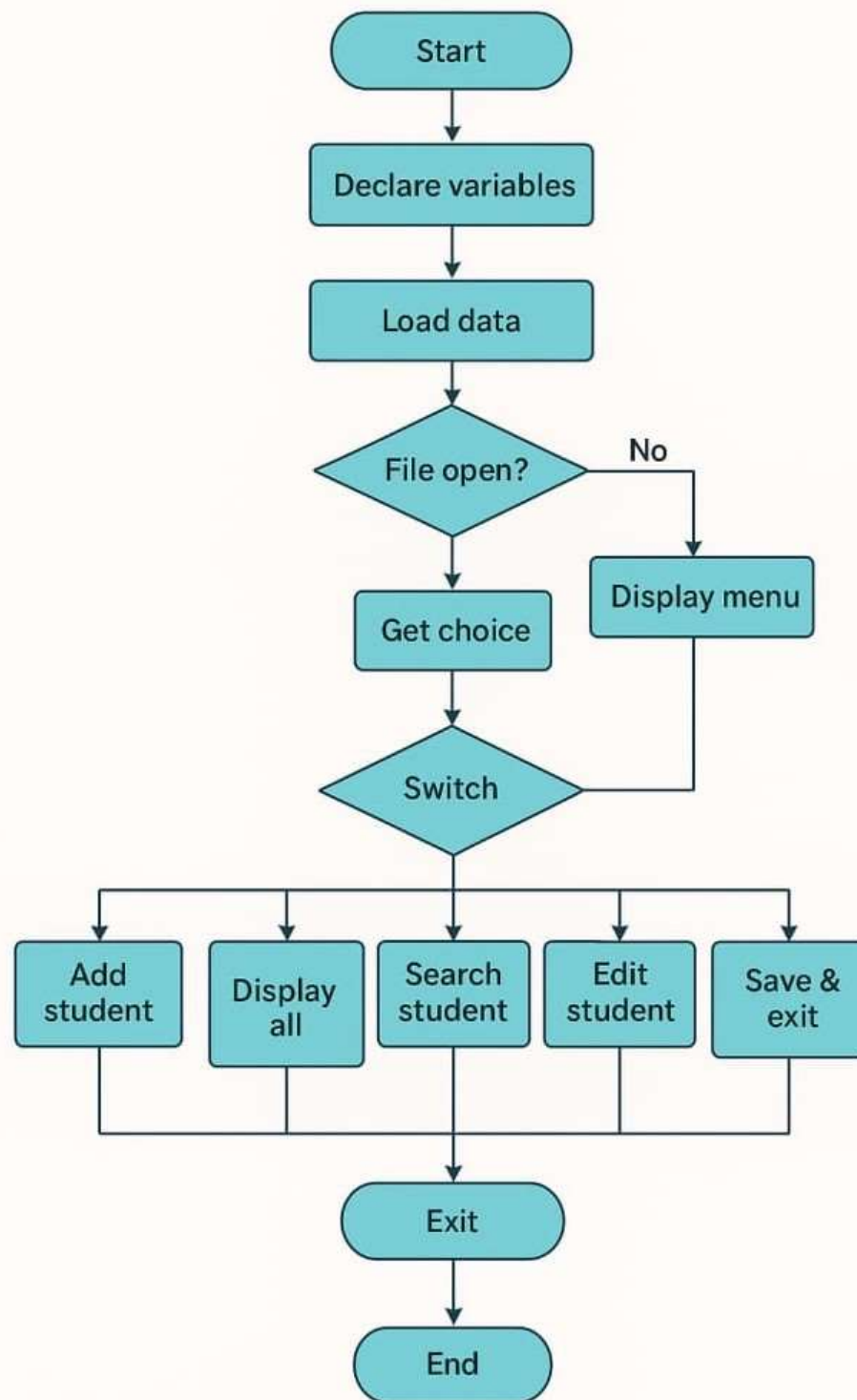
- To read student details from a file (`students.txt`) when the program starts.
- Read roll number, name, and marks
- Recalculate total, average, and grade.

4. Main Menu Module

main() acts as controller . It shows menu and routes to functions.

1. Create Student
2. Add Student Details
3. Calculate Result for a Student
4. Display All Student Details
5. Search Student
6. Modify Student Record
7. Exit

Flowchart



Code

```
#include <iostream>

#include <fstream>

#include <string>

using namespace std;


class Student {
private:
    int rollNo;

    string name;

    int marks[10];

    int total;

    float average;

    char grade;

    int n;

public:
    void inputDetails() {
        cout << "\nEnter Roll Number: ";
```

```
cin >> rollNo;

cin.ignore(1000, '\n');

cout << "Enter Name : ";

getline(cin, name);


total = 0;

cout<<"Enter number of subjects: ";

cin>>n;

for (int i = 0; i < n; i++) {

    cout << "Enter marks for subject " << i + 1 << ": ";

    cin >> marks[i];

    total += marks[i];

}

average = total / n;


if (average >= 90) grade = 'A';

else if (average >= 75) grade = 'B';

else if (average >= 60) grade = 'C';

else if (average >= 50) grade = 'D';

else grade = 'F';

}


void displayDetails() {

    cout << "\n----- STUDENT DETAILS ----- \n";

    cout << "Name : " << name << endl;

    cout << "Roll No : " << rollNo << endl;

    cout << "Total Marks : " << total << endl;
```



```
    cout << "Average : " << average << endl;

    cout << "Grade : " << grade << endl;

    cout << "-----\n";

}
```

```
int getRollNo() { return rollNo; }
```

```
void SaveToFile(ofstream &out) {

    out << rollNo << endl;

    out << name << endl;

    for (int i = 0; i < 10; i++)

        out << marks[i] << " ";

    out << endl;

}
```

```
bool readFromFile(ifstream &in) {

    if (!(in >> rollNo))

        return false;

    in >> name;

    total = 0;

    for (int i = 0; i < 10; i++) {

        in >> marks[i];

        total += marks[i];

    }

    average = total / float(n);
```

```
    if (average >= 90) grade = 'A';
```

```
    else if (average >= 75) grade = 'B';  
    else if (average >= 60) grade = 'C';  
    else if (average >= 50) grade = 'D';  
    else grade = 'F';  
    return true;  
}
```

```
void editDetails() {  
    cout << "\nEditing Student (Roll No: " << rollNo << ")\n";  
    cin.ignore(1000, '\n');  
    cout << "Enter new Name : ";  
    getline(cin, name);  
  
    total = 0;  
    cout<<"Enter number of subjects: ";  
    cin>>n;  
  
    for (int i = 0; i < n; i++) {  
        cout << "Enter marks for subject " << i + 1 << ": ";  
        cin >> marks[i];  
  
        total += marks[i];  
    }  
    average = total / float(n);  
  
    if (average >= 90) grade = 'A';  
    else if (average >= 75) grade = 'B';
```

```

        else if (average >= 60) grade = 'C';
        else if (average >= 50) grade = 'D';
        else grade = 'F';

        cout << "Record updated successfully.\n";
    }
};

int main() {
    Student s1[100];
    int count = 0;
    int choice;
    ifstream inFile("students.txt");
    if (inFile) {
        Student temp;
        while (temp.readFromFile(inFile)) {
            s1[count++] = temp;
            if (count >= 100) break;
        }
        inFile.close();
    }

    do {
        cout << "\n===== STUDENT GRADE MANAGEMENT SYSTEM =====\n";
        cout << "1. Add Student Details\n";
        cout << "2. Display All Student Details\n";
        cout << "3. Search Student by Roll Number\n";
    }
}

```

```
cout << "4. Edit Student Details\n";

cout << "5. Save & Exit\n";

cout << "Enter your choice: ";

cin >> choice;


if (choice == 1) {
    if (count < 100) {

        s1[count].inputDetails();

        count++;

        cout << "Student Added Successfully!\n";

    } else {

        cout << "Maximum student limit reached.\n";

    }
}

else if (choice == 2) {
    if (count == 0)

        cout << "No Student Records Found!\n";

    else

        for (int i = 0; i < count; i++)

            s1[i].displayDetails();

}

else if (choice == 3) {

    int r, found = 0;

    cout << "Enter Roll Number to search: ";

    cin >> r;
```

```

        for (int i = 0; i < count; i++) {
            if (s1[i].getRollNo() == r) {
                s1[i].displayDetails();

                found = 1;

                break;
            }
        }

        if (!found)

            cout << "Student Not Found!\n";
    }

    else if (choice == 4) {
        int r, found = 0;

        cout << "Enter Roll Number to edit: ";

        cin >> r;

        for (int i = 0; i < count; i++) {
            if (s1[i].getRollNo() == r) {
                s1[i].editDetails();

                found = 1;

                break;
            }
        }

        if (!found)

            cout << "Student Not Found!\n";
    }

    else if (choice == 5) {
        ofstream fout("students.txt");
    }

```

```
        for (int i = 0; i < count; i++)  
            s1[i].SaveToFile(fout);  
        fout.close();  
        cout << "Data Saved Successfully. Exiting...\n";  
    }  
    else {  
        cout << "Invalid Choice! Try Again.\n";  
    }  
  
} while (choice != 5);  
  
return 0;  
}
```

Output

Test Case 1:

```
===== STUDENT GRADE MANAGEMENT SYSTEM =====
1. Add Student Details
2. Display All Student Details
3. Search Student by Roll Number
4. Edit Student Details
5. Save & Exit
Enter your choice: 1

Enter Roll Number: 80
Enter Name : Gurleen Kaur
Enter number of subjects: 5
Enter marks for subject 1: 96
Enter marks for subject 2: 94
Enter marks for subject 3: 98
Enter marks for subject 4: 95
Enter marks for subject 5: 97
Student Added Successfully!
```

Test Case 2:

```
===== STUDENT GRADE MANAGEMENT SYSTEM =====
1. Add Student Details
2. Display All Student Details
3. Search Student by Roll Number
4. Edit Student Details
5. Save & Exit
Enter your choice: 2

----- STUDENT DETAILS -----
Name : Gurleen Kaur
Roll No : 80
Total Marks : 480
Average : 96
Grade : A
-----
```

Test Case 3:

```
===== STUDENT GRADE MANAGEMENT SYSTEM =====  
1. Add Student Details  
2. Display All Student Details  
3. Search Student by Roll Number  
4. Edit Student Details  
5. Save & Exit  
Enter your choice: 3  
Enter Roll Number to search: 96  
Student Not Found!
```

Test Case 4:

```
===== STUDENT GRADE MANAGEMENT SYSTEM =====  
1. Add Student Details  
2. Display All Student Details  
3. Search Student by Roll Number  
4. Edit Student Details  
5. Save & Exit  
Enter your choice: 1  
  
Enter Roll Number: 96  
Enter Name : Harleen Kaur  
Enter number of subjects: 5  
Enter marks for subject 1: 96  
Enter marks for subject 2: 94  
Enter marks for subject 3: 95  
Enter marks for subject 4: 92  
Enter marks for subject 5: 91  
Student Added Successfully!
```


Test Case 5:

```
===== STUDENT GRADE MANAGEMENT SYSTEM =====
```

1. Add Student Details
2. Display All Student Details
3. Search Student by Roll Number
4. Edit Student Details
5. Save & Exit

Enter your choice: 2

```
----- STUDENT DETAILS -----
```

Name : Gurleen Kaur

Roll No : 80

Total Marks : 480

Average : 96

Grade : A

```
----- STUDENT DETAILS -----
```

Name : Harleen Kaur

Roll No : 96

Total Marks : 468

Average : 93

Grade : A

Test Case 6:

```
===== STUDENT GRADE MANAGEMENT SYSTEM =====
```

1. Add Student Details
2. Display All Student Details
3. Search Student by Roll Number
4. Edit Student Details
5. Save & Exit

Enter your choice: 4

Enter Roll Number to edit: 69

Student Not Found!

Test Case 7:

```
===== STUDENT GRADE MANAGEMENT SYSTEM =====
1. Add Student Details
2. Display All Student Details
3. Search Student by Roll Number
4. Edit Student Details
5. Save & Exit
Enter your choice: 2

----- STUDENT DETAILS -----
Name : GUrleen Kaur Sran
Roll No : 80
Total Marks : 457
Average : 91.4
Grade : A
-----

----- STUDENT DETAILS -----
Name : Harleen Kaur
Roll No : 96
Total Marks : 468
Average : 93
Grade : A
-----
```

Test Case 8:

```
===== STUDENT GRADE MANAGEMENT SYSTEM =====
1. Add Student Details
2. Display All Student Details
3. Search Student by Roll Number
4. Edit Student Details
5. Save & Exit
Enter your choice: 5
Data Saved Successfully. Exiting...
```